

Advanced Data Analytics

Course orientation

Hye Hyun Ma, 2026. 9. 2.



How agents are transforming work

THE SHIFT TO AGENTIC AI: EVIDENCE FROM CODEX

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We analyze usage data from OpenAI's Codex tool to present large-scale evidence of how agentic AI technology, which can take actions on a user's behalf, changes how people work. We use an automated, privacy-protecting pipeline to contrast usage across three populations: external personal-account users, external organizational-account users, and workers within OpenAI. We find that agentic AI usage is growing rapidly: the number of active users has grown more than fivefold in the first half of 2026, with the most rapid increase occurring outside the initial audience of software developers. Uptake is uneven across contexts: within OpenAI, Codex usage is nearly universal and has largely replaced business usage of ChatGPT. We document a similar shift to agentic tooling outside OpenAI, particularly within organizations, although external adoption remains lower and more uneven. In addition to headline usage figures, we observe measures of sophistication, and find that a growing number of users have used Codex to change their workflows substantially. We find that more than 10% of users manage three or more concurrent Codex agents at some point each week and that 26.6% use skills, which allow users to share instructions for complex workflows. Alongside these changes in usage practices, request complexity has increased: since the start of the year, the share of individual Codex users who submit at least one request for a task estimated to require more than eight hours for an experienced human to complete has increased nearly tenfold. Concurrently, output has grown rapidly—in June 2026, the median OpenAI employee in a legal role generated 13 times more monthly output tokens across Codex and ChatGPT than they did in November 2025, while the median researcher generated more than 50 times as many. We conclude by discussing the implications of these patterns for productivity, job reorganization, and workforce restructuring.

- Among OpenAI workers, Codex largely replaced ChatGPT as the main interface for work-related AI use: as of June 11, 2026, Codex accounts for 99.8% of output tokens these workers generate across Codex and ChatGPT.

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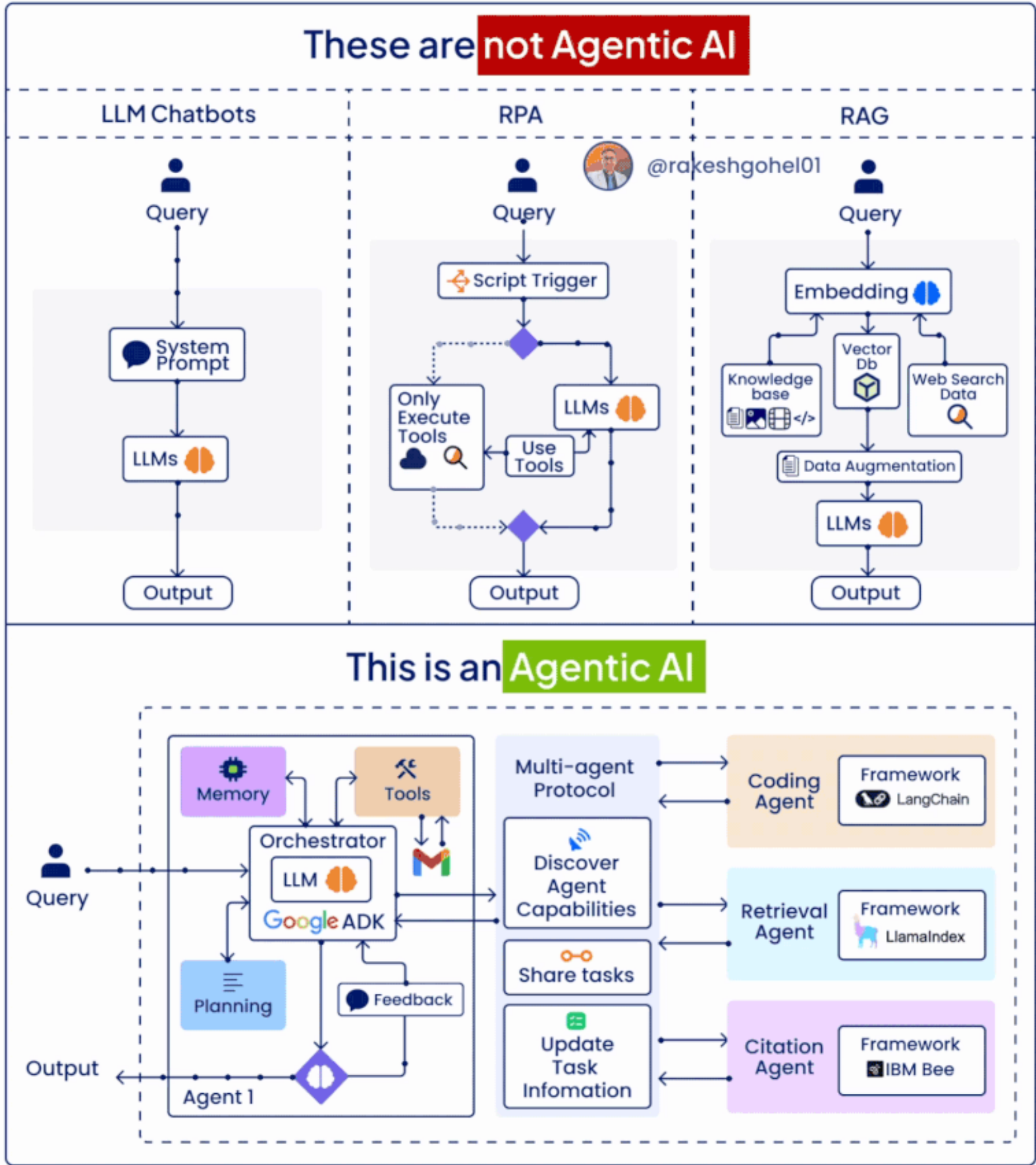
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What is actually an Agentic AI?



👨‍🍳: Find me a simple pasta recipe.

👩‍💼: Write a meeting summary.

👨‍💻: Suggest ideas for this poster.

👨‍🍳:
Plan dinner for four based on my preferences,
check what I already have,
order missing ingredients,
and adjust the plan if anything is unavailable.

👁️👁️ AI agents can even form a “society”

Generative Agents: Interactive Simulacra of Human Behavior

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Figure 1: Generative agents are believable simulacra of human behavior for interactive applications. In this work, we demonstrate generative agents by populating a sandbox environment, reminiscent of The Sims, with twenty-five agents. Users can observe and intervene as agents plan their days, share news, form relationships, and coordinate group activities.

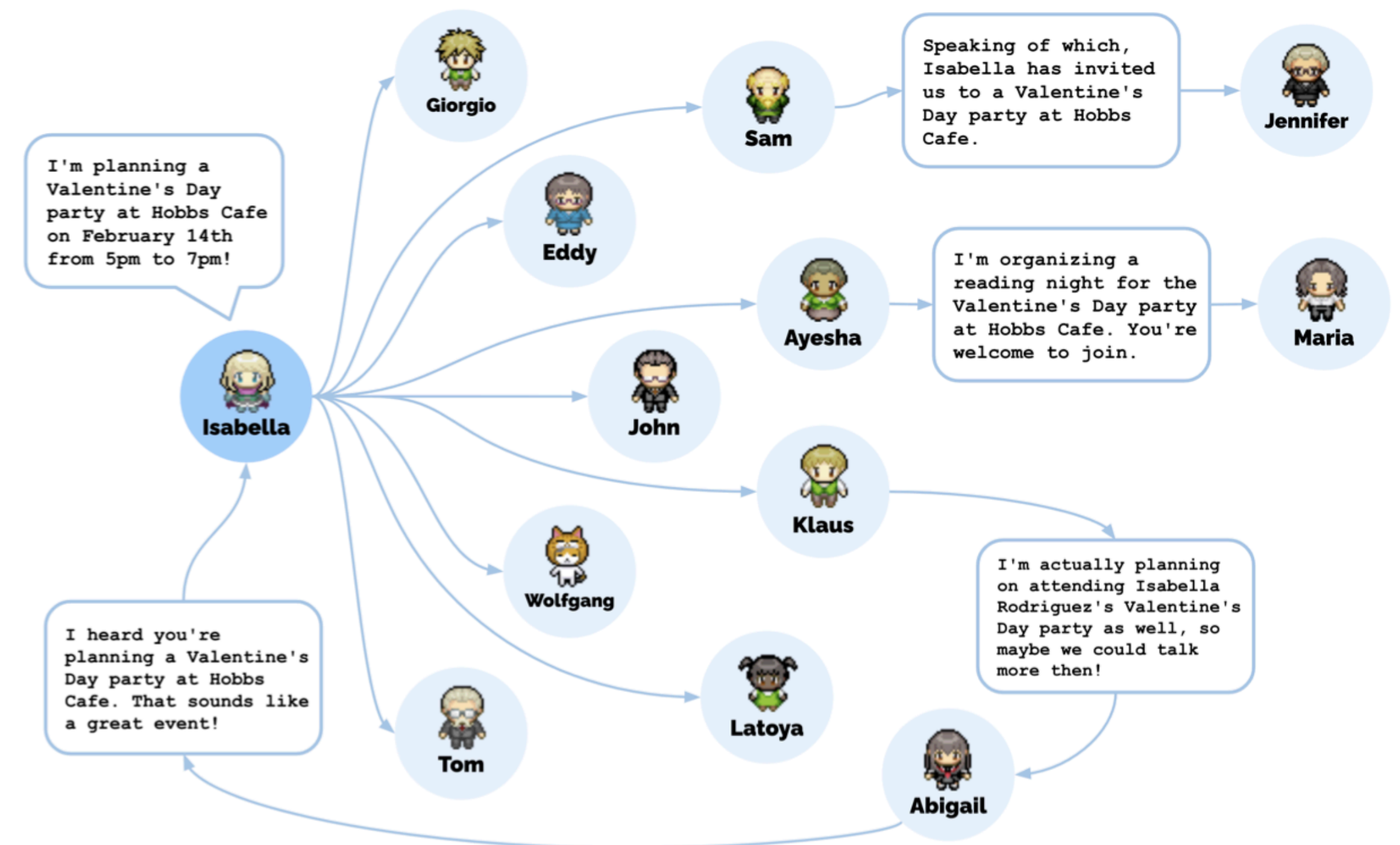


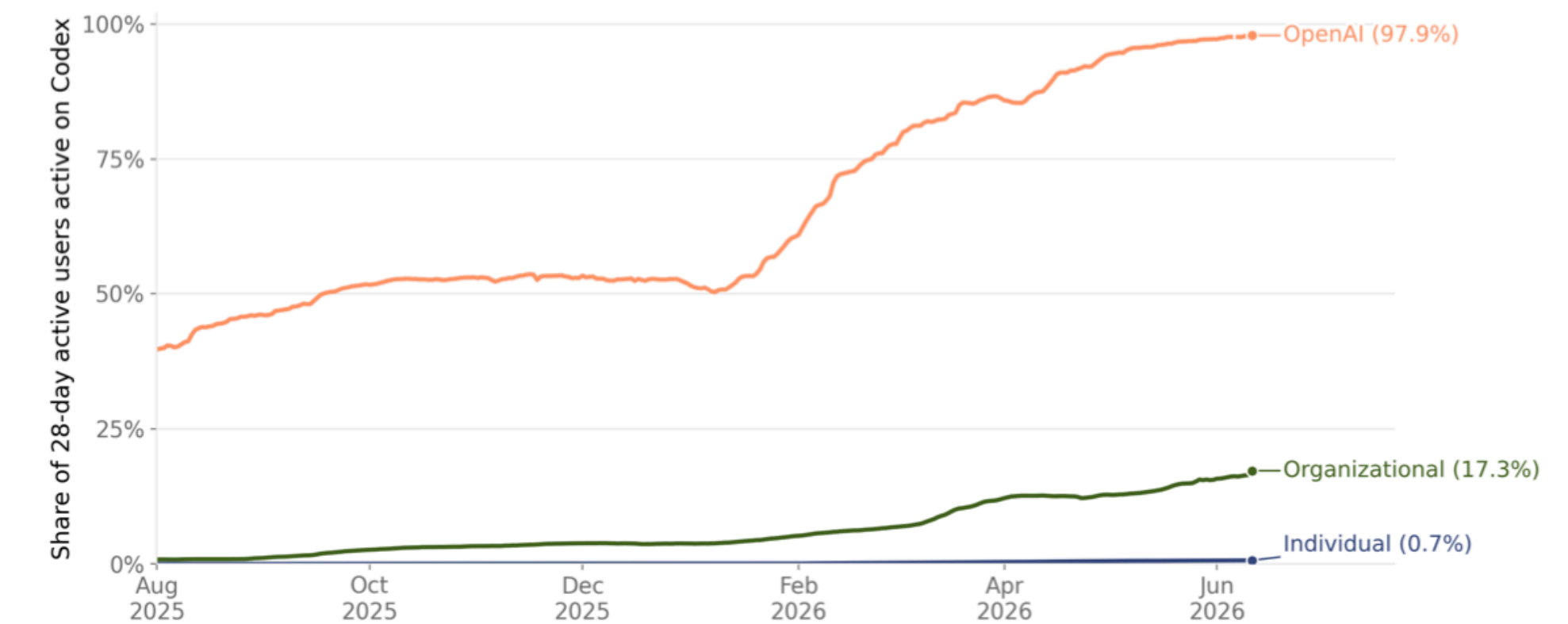
Figure 9: The diffusion path for Isabella Rodriguez’s Valentine’s Day party invitation involved a total of 12 agents, aside from Isabella, who heard about the party at Hobbs Cafe by the end of the simulation.

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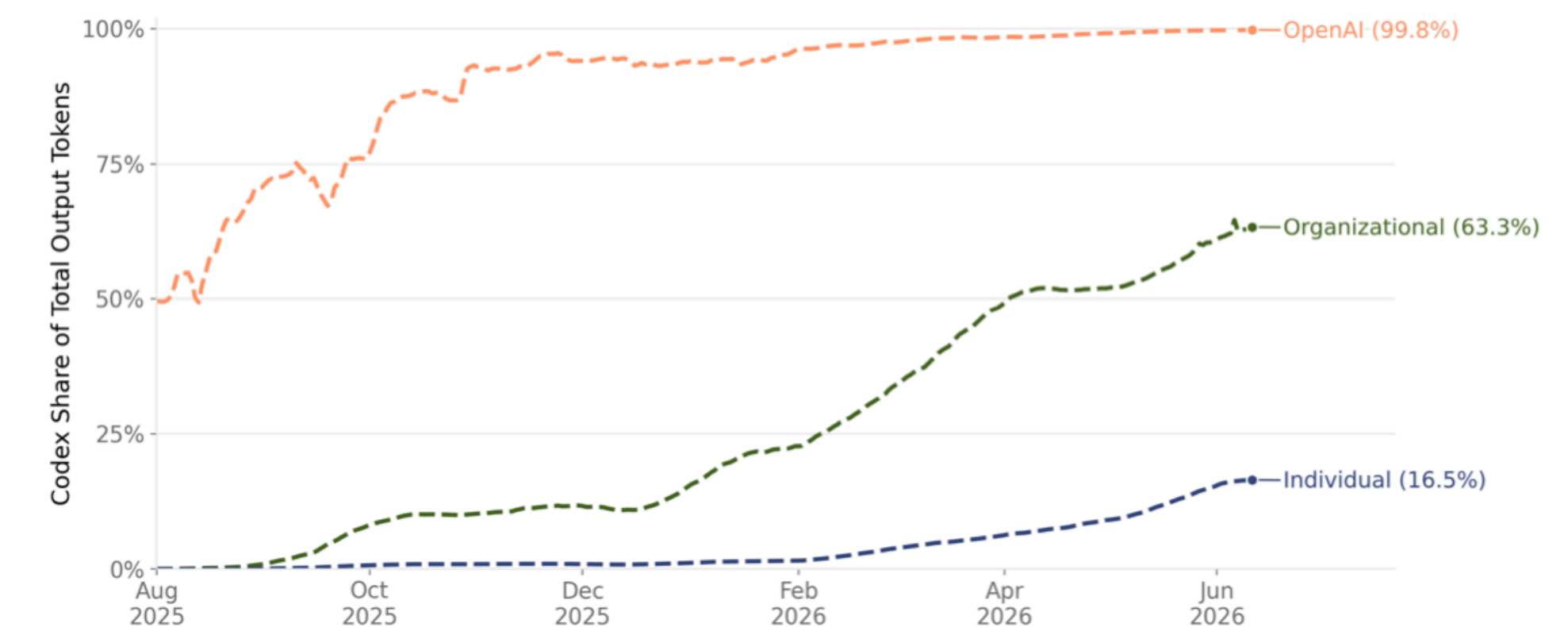
Four trends over the past year

- People use Codex for longer-horizon work.
- Codex became the primary AI tool for every department at OpenAI.
- Non-developer adoption grew especially rapidly, outpacing developer adoption.
- Codex enabled OpenAI workers to do tasks outside their job description.

Fig. 1: Codex usage relative to ChatGPT, by user population



(a) Share of active users using Codex

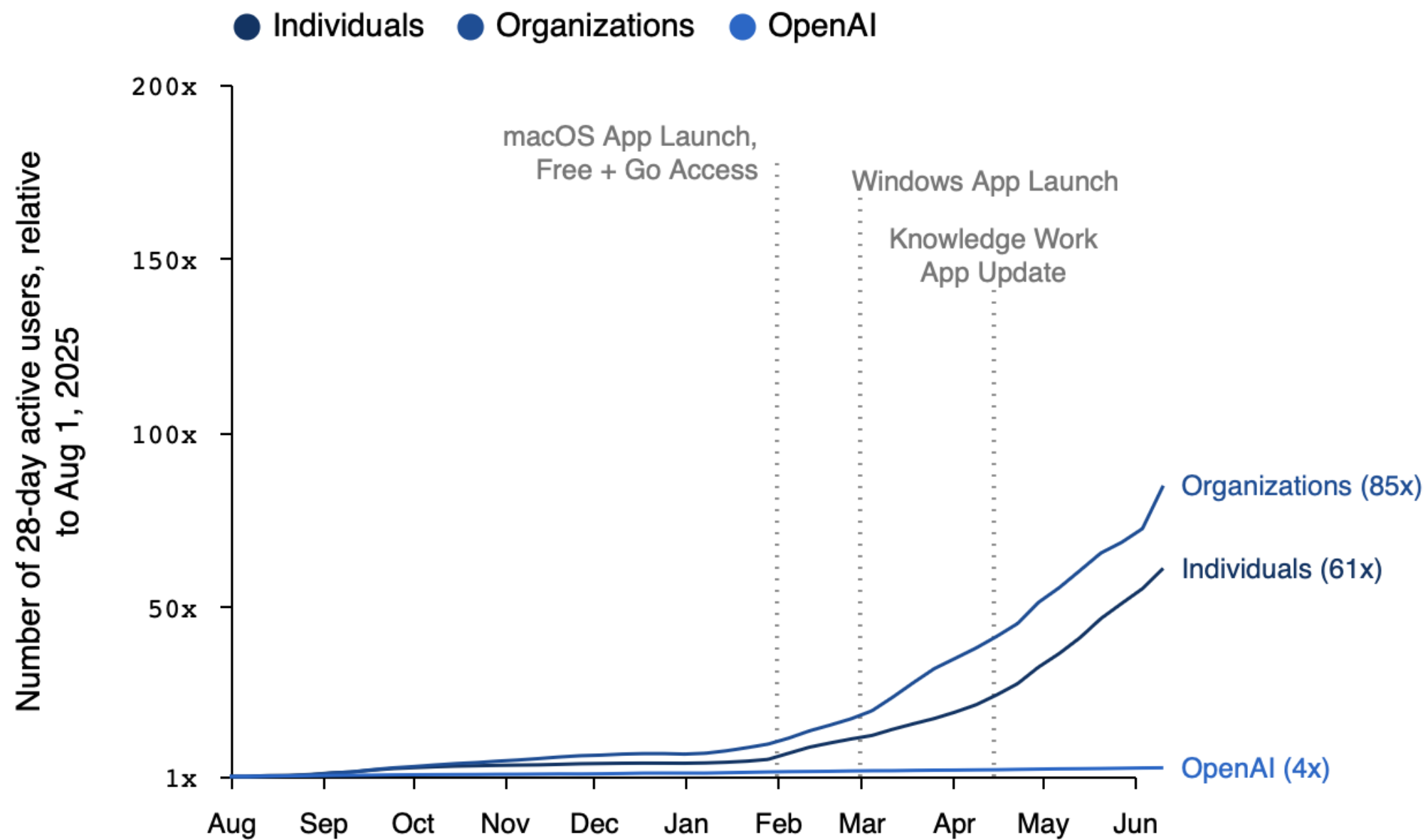


(b) Share of output tokens from Codex

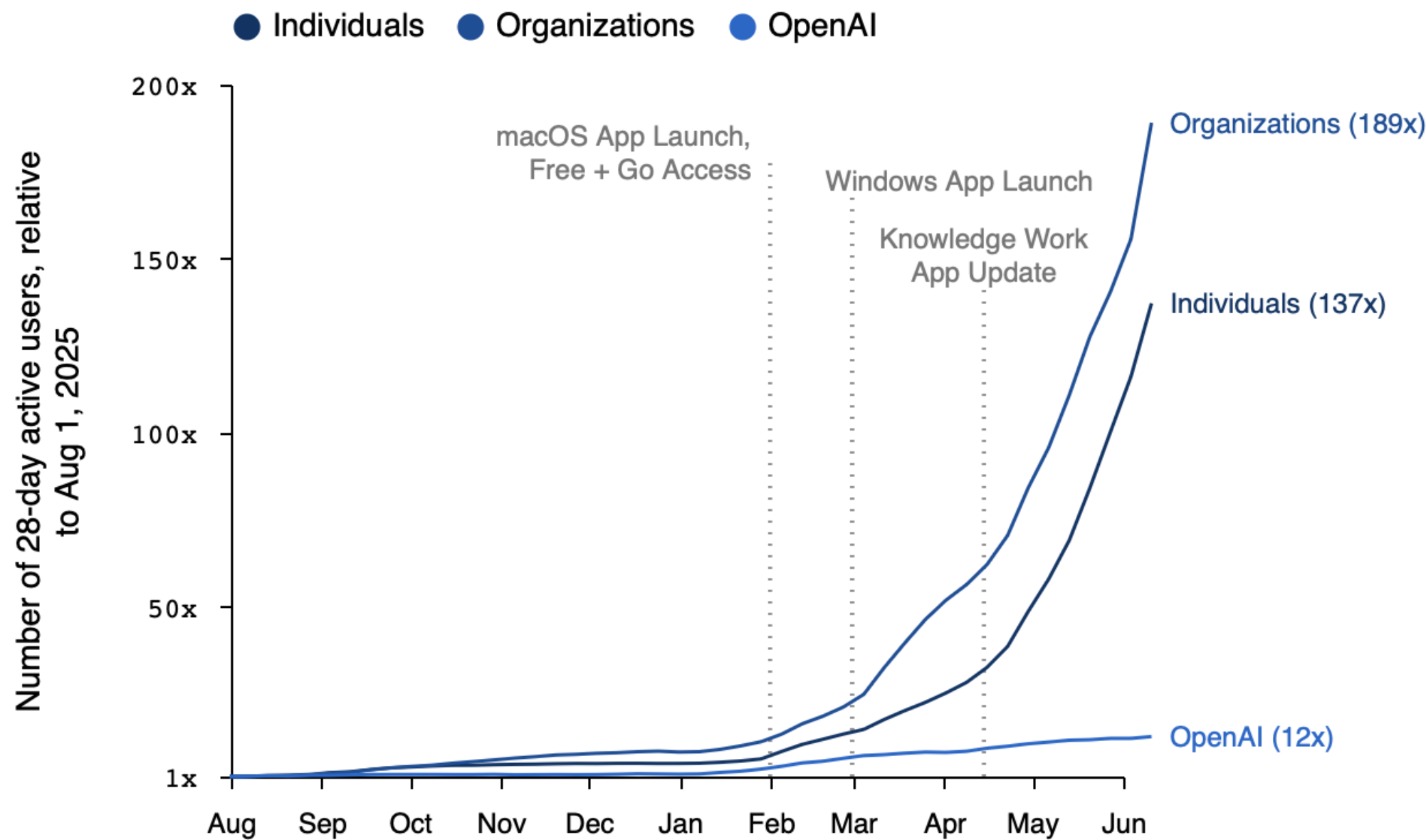
Panel A shows the share of users active on either ChatGPT or Codex during the preceding twenty-eight days who were active on Codex at least once during that period. Panel B shows output tokens produced on Codex as a share of output tokens produced on either Codex or ChatGPT during the preceding twenty-eight days. We compare individual users, organizational users, and OpenAI workers.

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Developer



Non-developer



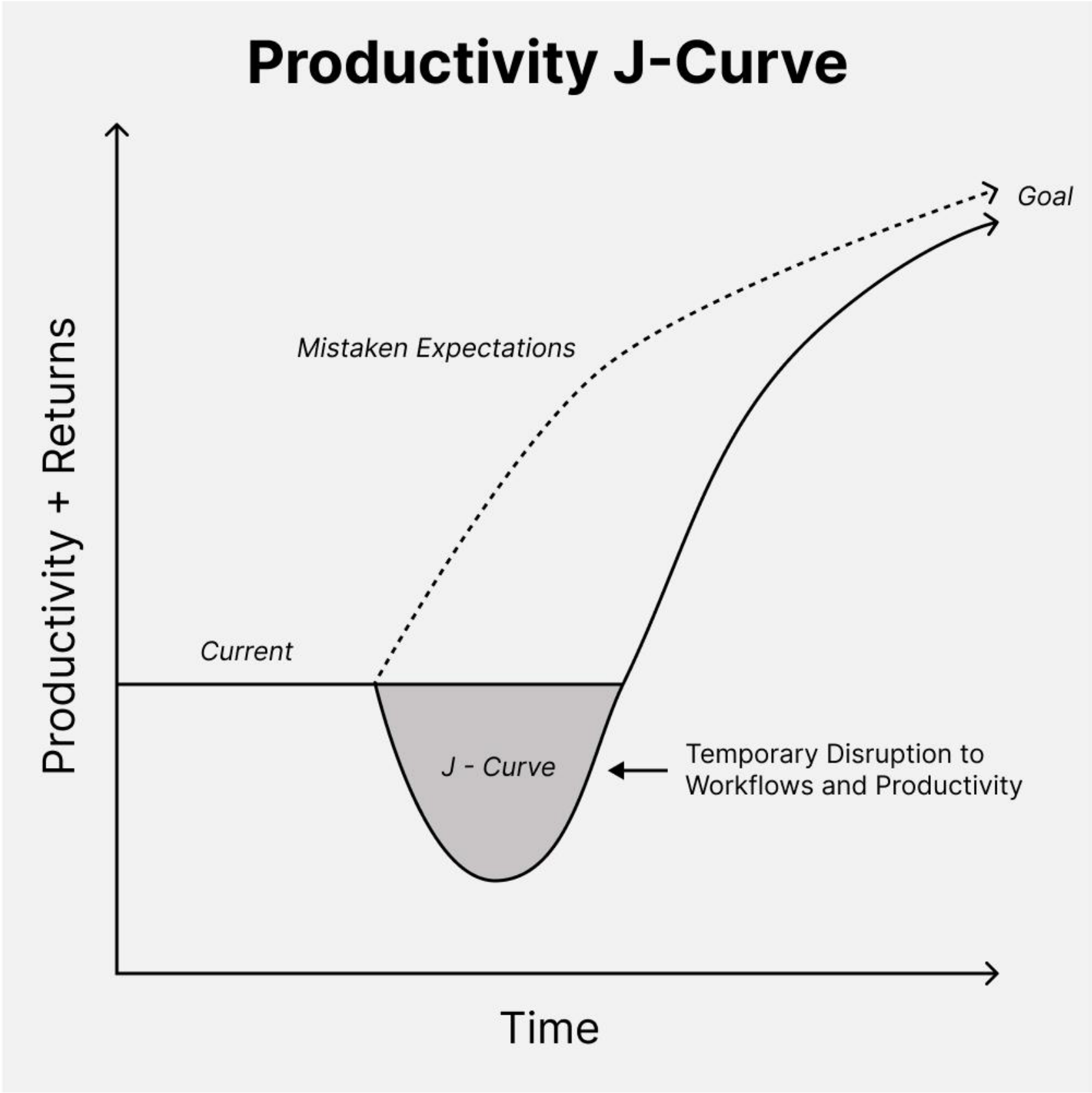
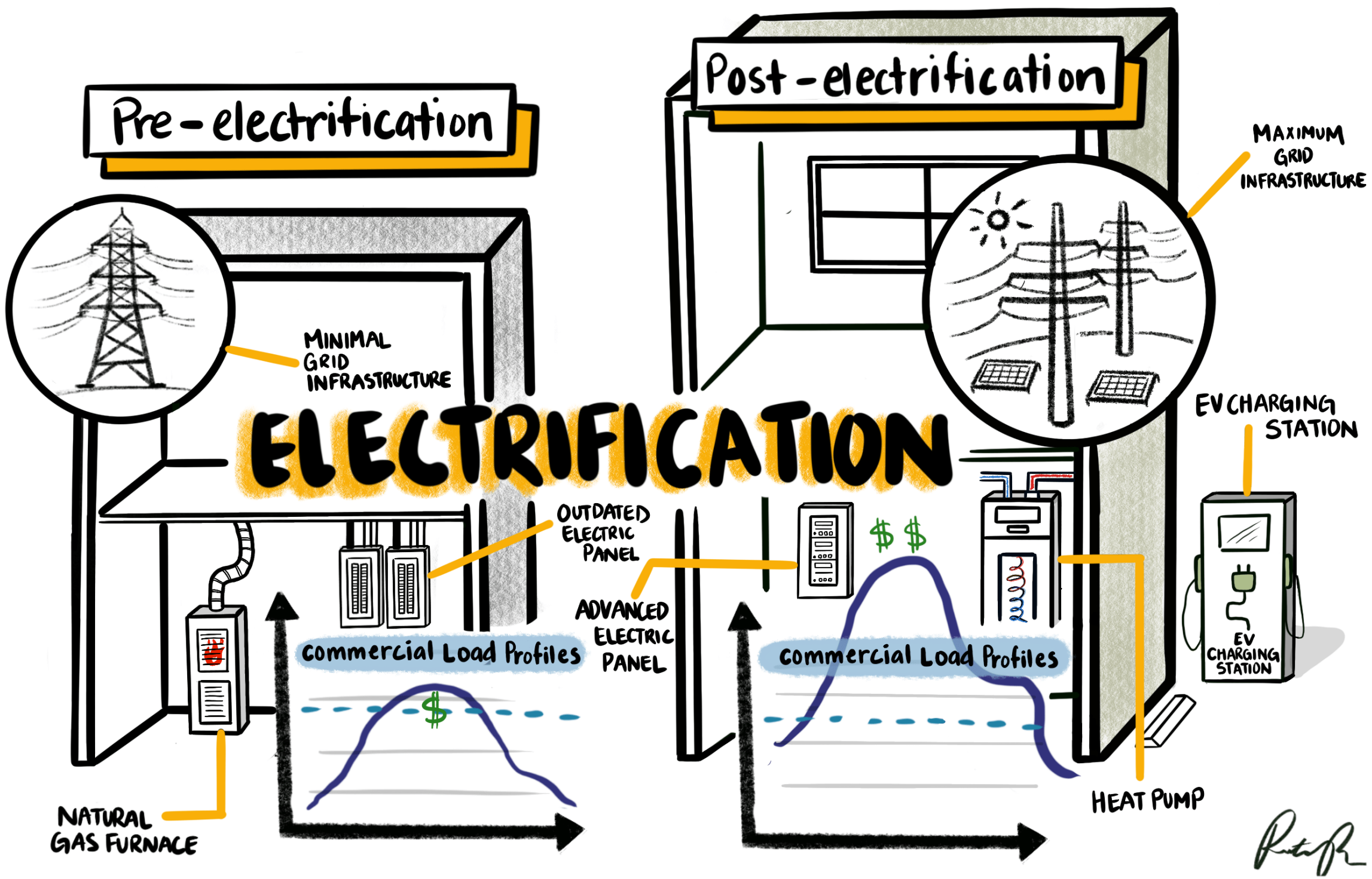
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Occupation vs. work done with Codex



Consolidated inferred department	Work category				
	Engineering / Coding	Data Analysis	Financial Analysis	Knowledge Work	Other
Engineering	72%	4%	1%	18%	5%
Data Science / Research	51%	10%	0%	30%	9%
Finance / Biz Ops	31%	9%	16%	34%	10%
Product / Marketing / Ops	25%	3%	7%	51%	15%
Other	50%	7%	2%	38%	4%

How agents are transforming work



Is AI adoption always smooth?

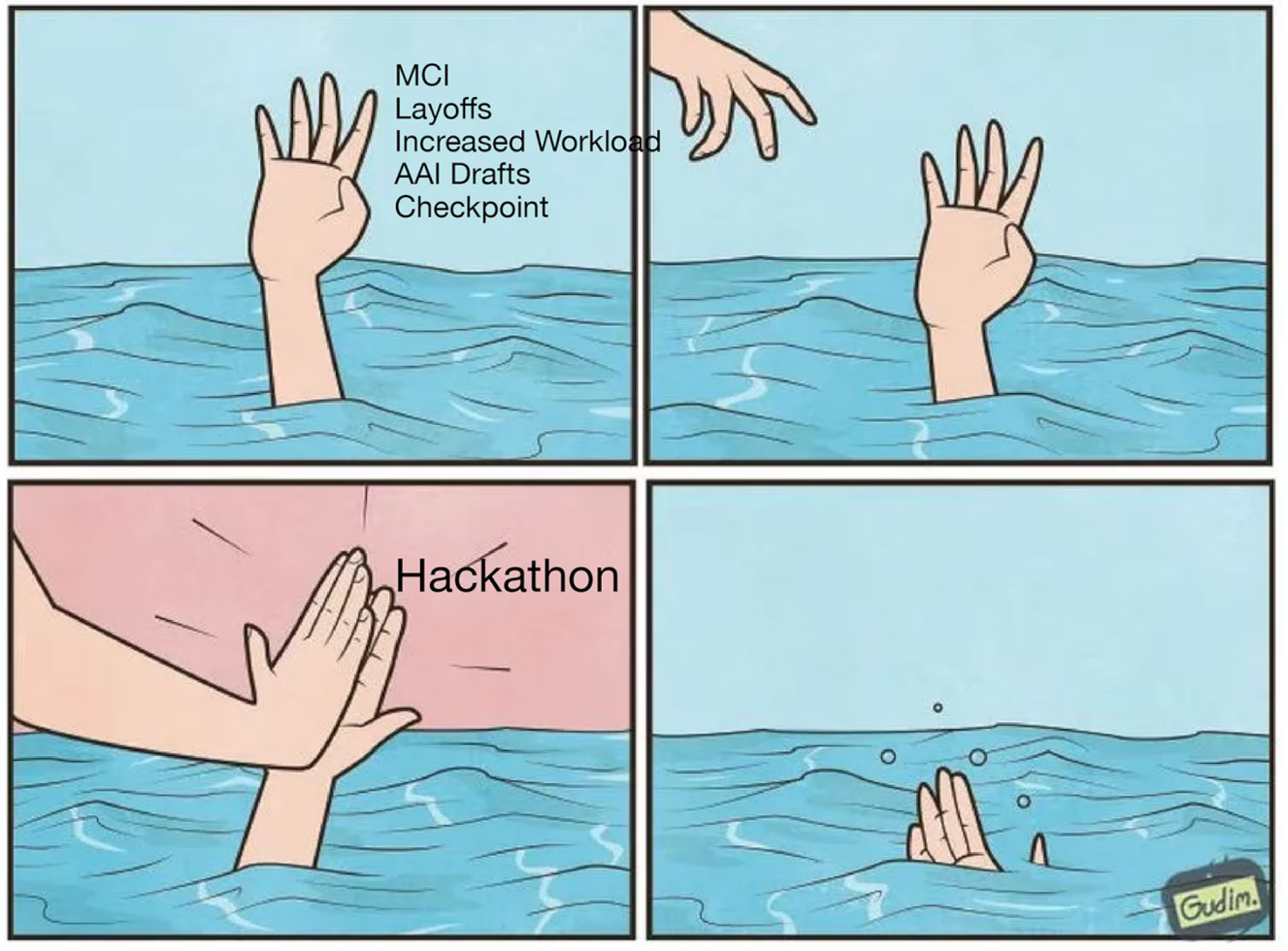
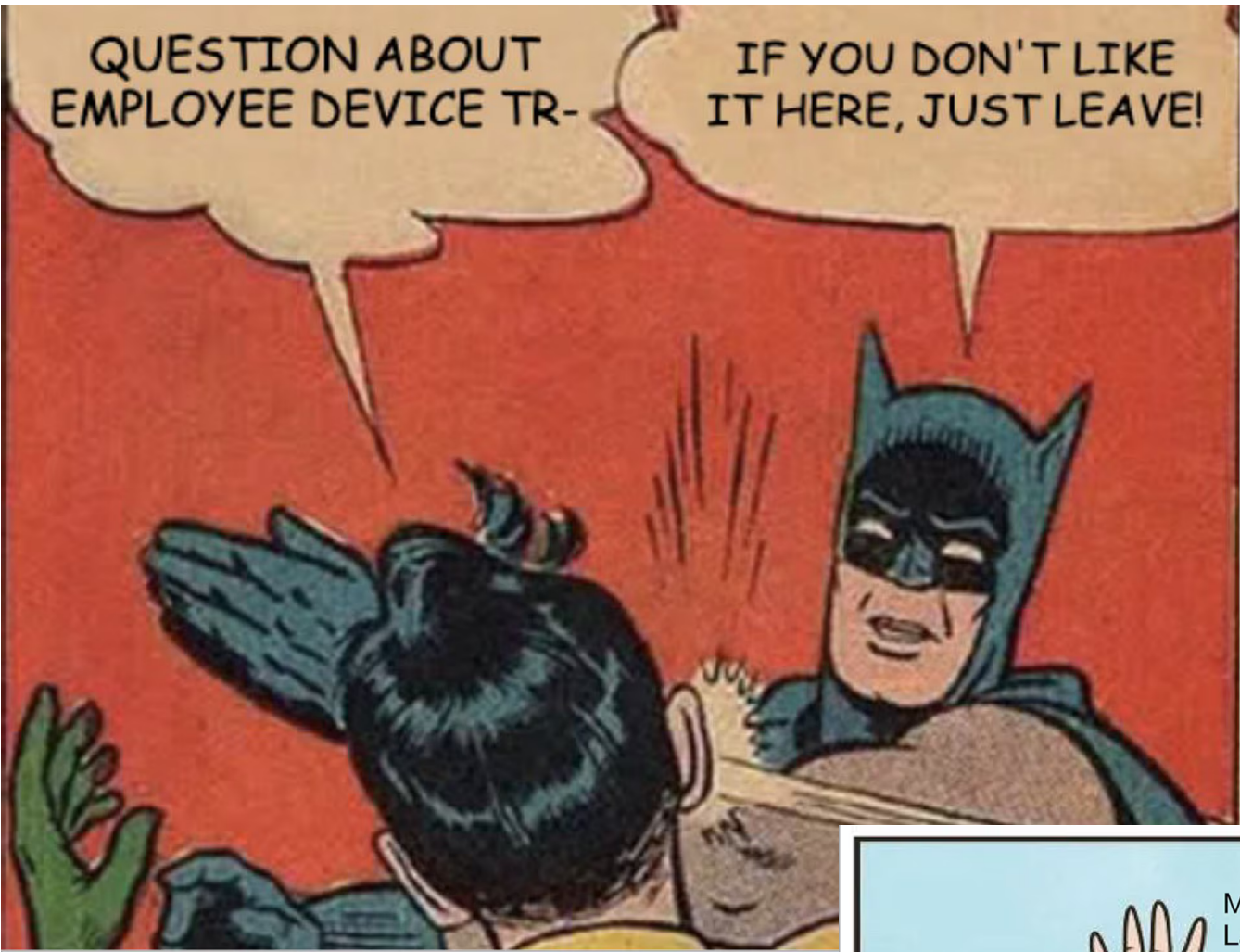
A REUTERS SPECIAL REPORT

Mark Zuckerberg had a bold plan to replace Meta staff with AI. Here's how it imploded.

To make its workforce “AI native,” Meta explored slashing the size of many teams across the company by as much as 60% in two waves, internal documents show. Staff revolted - and hours before a big layoff in May, the CEO called off planning for further cuts.

By **Katie Paul**

August 26, 2026 7:00 PM GMT+9 · Updated August 26, 2026



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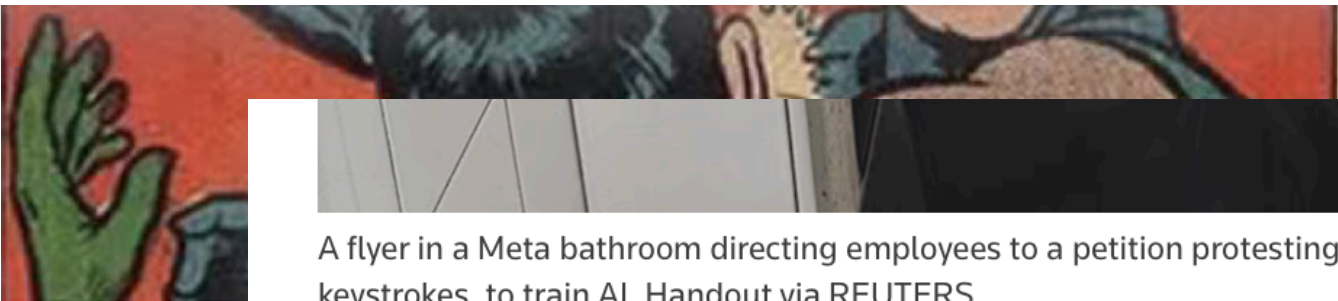
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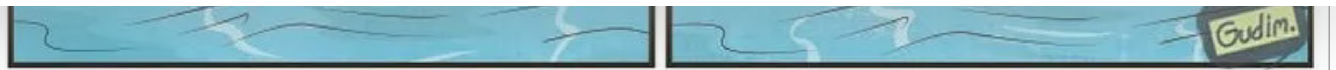


Amid the rebellion, other internal data indicated that the tech at the heart of the plan wasn't delivering. Employees' use of AI had resulted in a vast increase in the code they generated, but with questionable impact on productivity, according to internal posts seen by Reuters. For instance, code changes made to the internal software platforms and infrastructure employees use on the job were up 220% year-over-year, according to a post by Bosworth in early June. But changes that led to new or upgraded features reaching Meta users were only up 36%.



As early as March, infrastructure teams were also warning of “reliability warning signs” caused by the AI coding surge, according to an internal post. Another post, in April, said that unchecked AI agents were performing “large-scale, disruptive actions that humans are unlikely to execute.” The result: Major technical and security incidents, such as service disruptions and possible data leaks, spiked 40% from the previous year, with the time staffers had to spend “firefighting” them up 70%, according to the internal posts.

The public saw a glimpse of the problem in early June after hackers exploited Meta's new AI-powered customer support bot to gain access to high-profile Instagram accounts, including one for the dormant Obama White House page.



AI adoption is not always smooth



What We'll Be Doing This Semester

- <https://docs.google.com/document/d/1HEQgB5-m247bci3iula5TQj5Ra9vk-i5D142Zo1Z-5k/edit?usp=sharing>